Anthony Baez

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Education

Massachusetts Institute of Technology — Cambridge, MA
Masters of Engineering in Artificial Intelligence and Decision Making — *2025 – Present*

Massachusetts Institute of Technology — Cambridge, MA
Bachelor of Science in Artificial Intelligence and Decision Making — *2021 – 2025*

Selected Coursework

Representation, Inference, and Reasoning in AI; Advances in Computer Vision; Quantitative Methods for Natural Language Processing; Optimization Methods; Diffusion Models: Theory to Practice; Deep Learning; Multiagent Learning

Research

MIT Media Lab — June 2025 – Present
Graduate Researcher — *Cambridge, MA*

- Developed validation methods and interfaces predicting LLM behavior using linear representations for customized system prompts and multi-turn interactions
- Created framework to decompose representations of sycophancy, can be extended to other behaviors
- Advised by Prof. Pat Pataranutaporn

MIT Research Laboratory of Electronics — Sept 2023 – May 2025
Undergraduate Researcher — *Cambridge, MA*

- Developed novel projection methods for guaranteeing linear and quadratic integrals in Physics-Informed Neural Networks
- Evaluated projection on multiple PDE benchmarks
- Advised by Prof. Luca Daniel

Universitat Pompeu Fabra NLP Research Group — June 2023 – Aug 2023
Visting Researcher — *Barcelona, Spain*

- Improved upon state-of-the-art benchmark in lexical simplification by supervised finetuning LLaMA
- Implemented data and model parallelism to finetune multi-GPU node
- Advised by Prof. Horacio Saggion

MIT Lab for Computational Physiology — June 2022 – Aug 2022
Undergraduate Researcher — *Cambridge, MA*

- Applied contrastive learning techniques to different classes of machine learning models to improve detection of cardiac arrhythmia for use in bedside monitors
- Adapted model to different ECG waveform datasets, conducted dataset and error analysis to improve models

Publications

Dissociating the Internal Representations of Sycophancy in LLMs 2026
***Anthony Baez**, Sheer Karny, and Pat Pataranutaporn*
Under Review

Multi-Turn Neural Transparency: Surfacing Neural Activations Improves User Calibration to LLM Behavioral Drift 2026
Sheer Karny, **Anthony Baez***, Pat Pataranutaporn (*equal contribution)*
Under Review

Neural Transparency: Mechanistic Interpretability Interfaces for Anticipating Model Behaviors for Personalized AI 2025
Sheer Karny, **Anthony Baez***, Pat Pataranutaporn (*equal contribution)*
Accepted to ACM Conference on Intelligent User Interfaces 2026

Guaranteeing Conservation of Integrals in Physics-Informed Neural Networks 2025
***Anthony Baez**, Wang Zhang, Ziwen Ma, Subhro Das, Lam M Nguyen, and Luca Daniel*
Submitted to ICLR 2026

Guaranteeing Conservation Laws in Physics-Informed Neural Networks 2024
***Anthony Baez**, Wang Zhang, Ziwen Ma, Subhro Das, Lam M Nguyen, and Luca Daniel*
Data-driven and Differentiable Simulations, Surrogates, and Solvers Workshop @ NeurIPS 2024

LSLlama: Fine-tuned LLaMA for Lexical Simplification 2023
***Anthony Baez** and Horacio Saggion*
Second Workshop on Text Simplification, Accessibility and Readability @ EMNLP 2023

Experience

DeepAI June 2024 – Aug 2024
Machine Learning Research Intern *San Francisco, CA*

- Created Python package that distributes workloads to decentralized GPU network
- Fine-tuned open source text-to-video model on mixed real-synthetic dataset to improve spatial understanding

Talks

Google Lightning Talk April 2026
AHA Members Week Showcase October 2025